Gap analysis and remediation plan: papers 01 and 02 for Trials

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Scope. This note assesses how far paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rm`
and paper 02 (`analysis/report/02-carryover-sensitivity/report.Rmd`)
are from being ready to submit to *Trials* (BioMed Central), and lays out a sequenced plan to close the gap. It assumes each paper's stated goal is the actual submission goal:

- **Paper 01's goal:** investigate defects in Hendrickson's published data-generating process (DGP) and recommend an improved DGP (the refined process) as the reference for aggregated N-of-1 simulation studies of this kind.
- **Paper 02's goal:** investigate candidate analysis specifications for the carryover problem and recommend one (currently Exposure-weighted, `G3`/`G8`, or possibly `G4`/`G9`) for use with that DGP.

Evidence labels follow the project convention: *verified* (observed by running code or inspecting output this session), *inspected* (read from source code or the manuscript text), *inferred* (consistent with surrounding material but not directly checked), and *unverified* (not checked at all). Items below are inspected/verified during this session's review unless marked otherwise.

A caveat on the journal-specific checklist in Part 3: I was not able to retrieve *Trials*' current Instructions for Authors page during this session (it redirected to an authenticated Springer login), so the journal-specific items are from general knowledge of BMC *Trials* practice and should be confirmed against the live page before submission.

Part 1: Paper 01 (DGP paper)

Where it stands

Paper 01 largely achieves its stated goal. The refined process is fully specified (Table `tab:spec`), matches the project's canonical definition in `docs/series/ch10-refined-process.md` exactly, and the four defect-and-repair pairs (step vs. graded coupling, compound symmetry vs. separable AR(1), compounding vs. anchored carryover, silent repair vs. stop-on-non-PD) are each documented with a closed-form or code-level explanation. The

Recommendations and Conclusions sections state the refined process as the paper's proposed reference DGP. This is a coherent, complete piece of work on its own terms.

Gaps against a Trials submission

1. **Missing headline result the rest of the project cites as coming from this paper.** [verified] The project's own `CLAUDE.md` and `NOTATION.md` Part 2 cite a matched-N (N=70, Hybrid) relative power loss under carryover of 15.4% (covariance moderation) versus 6.9% (mean moderation) as "paper 01 results." That comparison does not currently exist anywhere in `report.Rmd`. It survives only in an archived pre-restatement draft (`archive/report-2026-10-07-pre-restatement.Rmd`) and a review document, after the manuscript's scope was narrowed on 2026-10-07 to "specification of the compendium's reference DGP." Three other project documents point to a result that is not in the paper. This is the single highest-priority item: a referee who checks the cited figure against the manuscript will not find it.

2. **The package does not implement its own top recommendation.** [inspected, confirmed this session] `R/generateData.R` still silently repairs non-positive-definite covariance matrices via `corpcor::make.positive.definite()` rather than stopping, which is recommendation #1 in the paper's own Recommendations list ("Always apply the corrections: stop on non-positive-definite matrices…"). The paper's Appendix B code patches presumably implement the stop behavior for the paper's own analysis scripts, but the shipped package default does not. A *Trials* reviewer who reads the code-availability statement and checks the repository will find the package contradicting its own recommendation. Needs either a package-level fix or an explicit, visible statement in the paper that the stop behavior is implemented at the analysis-script level only, pending a package change.

3. **Companion-paper citations need a fabrication check.** [unverified] `report.Rmd` cites `@pmsimstats_dgp_architectures2026` and other in-house companion references. Given that this portfolio's referee-review campaign found fabricated-appearing content (invented seeds, unimplemented methods reported as implemented) in at least three other workspaces this year, every self-citation to an in-house companion paper or dataset should be checked for existence and content match before submission, not assumed correct because it is internal.

4. **Correction (2026-10-21): this item was wrong.** The original version of this gap analysis claimed the paper never names ADEMP explicitly. On a fuller read, `report.Rmd`'s "Simulation study design" section (around line 437) states plainly: "We designed and report the simulations following the

ADEMP structure of Morris et al. [@morris2019simulation]: aims, data-generating mechanisms, estimands, methods and performance measures," with Table `tab:ademp` immediately after. This gap does not exist; no action is needed. Left here, struck through in substance rather than deleted, so the correction is visible rather than silently erasing the earlier error.

5. **Parameter calibration is disclosed as a limitation but not mitigated.** [inspected] The Limitations section already states plainly that Gompertz, correlation, and coupling parameters are not estimated from real trial data. This is honest and adequate as a limitation statement, but a *Trials* reviewer may ask for at least a sensitivity range or a cited external plausibility argument (e.g., prazosin pharmacokinetic half-life literature) rather than a bare disclosure. Low priority relative to items 1-3.

Plan

Step	Action	Depends on	Status
1	Decide: reinstate the matched-N architecture-loss section in paper 01 (rerun if the archived numbers are stale), or formally redirect `CLAUDE.md/NOTATION.md/ch07-architectures.md` to cite wherever that analysis now belongs. This is a scope decision, not a writing task, and blocks everything else that treats "paper 01 results" as settled.	none	**Done (2026-10-21).** User decided: redirect. The 15.4%/6.9% figure is out of scope for paper 01, whose goal is the covariance-moderation DGP, not the architecture contrast.

Step	Action	Depends on	Status
2	If reinstating: pull the matched-N comparison and z-statistics from `archive/report-2026-10-07-pre-restatement.Rmd`, verify the numbers against a fresh run under the current refined-process code (package and parameters may have drifted since 2026-10-07), and reintegrate as a results section.	1	**Not applicable.** Redirect was chosen over reinstatement.

Step	Action	Depends on	Status
3	Resolve the non-PD stop-vs-repair mismatch: either patch `R/generateData.R` to add a `stop_on_nonpd` argument used by default in the paper's own driver scripts, or add an explicit statement in Methods naming this as a package-vs-analysis-script distinction.	none	**Done (2026-10-21).** User decided: patch the package. `buildSigma()`/`generateData()` now accept `stopOnNonPD` (default `FALSE`, backward compatible); when `TRUE` it raises an error instead of repairing, at both the explicit PD check and the previously-silent Cholesky-fallback repair. Tests

Step	Action	Depends on	Status
4	Verify every in-house companion citation (`@pmsimstats_dgp_architectures2026`, etc.) resolves to an existing, content-matched document.	none	**Done for paper 01 (2026-10-21).** Every `\@`-citation in `report.Rmd` (excluding bookdown's `\@ref()` cross-references, a false positive in the original check) resolves to a real `references.bib` entry; no fabricated citations found in paper 01

Step	Action	Depends on	Status
5	Add an explicit ADEMP-conformance statement in Methods, mapping the paper's existing table structure onto ADEMP's five elements by name.	none	**Not needed — already present.** See the correction to gap item 4 above; this was an error in the original gap analysis.

Step	Action	Depends on	Status
6	Full read-through against *Trials*' current Instructions for Authors (word limit, structured abstract, declarations, data/code availability statement, reporting-guideline checklist) once confirmed live.	1-5	**Blocked.** `trialsjournal.biomedcentral` submission-guidelines page redirects to an authenticated Springer login; could not be fetched this session. Needs a manual check against the live page.

Paper 01's remaining gaps (steps 1, 3, 4, 5) are resolved as of 2026-10-21. Step 6 (the journal-specific read-through) is the only item still open, and it is blocked on access to *Trials*' live Instructions for Authors rather than on any further work in this repository.

Part 2: Paper 02 (analysis-method paper)

Where it stands

Paper 02 is a substantial, well-scoped comparison: nine analysis specifications (`G1`-`G9`), explicit scoping rationale for what was and was not tested, a clear negative result (the lagged-treatment term does nothing), a higher-precision paired-McNemar confirmation of the headline ranking, and an explicit (if incomplete) check of the post-selection inference risk in its data-driven half-life specification. This is further from submission-ready than paper 01, for reasons specific to this session's review plus the DGP inheritance problem raised by the user's prompt ("02 should use the canonical DGP").

Gaps against a Trials submission

1. **The simulation driver does not fully implement the refined process paper 01 is recommending as the project's reference DGP.** [verified this session, see prior turns] Checked against the five refined-process properties:

   - Graded coupling: present, correctly implemented.
   - Anchored carryover: present, correctly implemented.
   - Separable AR(1) at $\rho = 0.7$: present, but the **cross-factor correlation constant is numerically wrong**. The canonical form is a single $c_1 = 0.2$ scaled by $\rho^{|w_t - w_s|}$ at every lag; the driver (`simulation-core.R:444,839`) instead uses two different constants, 0.2 at same-timepoint and $0.1 \times \rho^{\text{lag}}$ at every other lag. This is a real numeric deviation from the documented spec, not an undocumented-but-correct case.
   - Stop on non-PD matrices: absent. The driver calls with `make_positive_definite = TRUE`, i.e., silent repair, matching the published process, not the refined one.
   - Randomized path allocation: absent. Patients are split across paths deterministically (`floor(N/n_paths)` plus remainder to the first paths), with no `sample()` call anywhere in the driver.

   This means paper 02's simulations are not run under the DGP that its own companion paper is about to recommend as the project's reference process. This is the direct answer to "02 should use the canonical DGP": it does not, on two of five properties, and is numerically off on a third.

2. **Stale architecture terminology throughout.** [inspected] The project retired the "Architecture A/B" labels for covariance-moderation/mean-moderation on 2026-09-30-ish per `NOTATION.md` Part 3, but paper 02 still uses "Architecture B (covariance moderation) only" in its DGP-architecture table row and in multiple figure captions ("Architecture A",

"Architecture B"). A referee reading both papers side by side, as a *Trials* reviewer handling companion submissions likely would, will notice the terminology mismatch immediately.

3. **No explicit DGP-specification section.** [inspected] Unlike paper 01's Table `tab:spec`, paper 02's "Data-generating mechanism" section (lines 505-575) never names "refined process," "anchored," "c_1," "graded coupling," or "randomized allocation." Even the two properties that are correctly implemented (graded coupling, anchored carryover) are described without using the project's own established vocabulary for them, so a reader cannot tell from the text alone whether this is the same DGP as paper 01's recommendation.

4. **The data-driven half-life specification's Type I error claim is under-supported and inconsistent with the project's own closer look at the same question.** [verified this session] Block S11 checks `G4` (AIC-selected half-life, uncalibrated) and `G9` (`G4`+CR2) for Type I error inflation at one cell (Hybrid, N=70, true $t_{1/2} = 1.0$ week, $n_{sim} = 2000$) and finds none (0.038 and 0.052 respectively), explicitly noting this "does not generalize beyond the single cell tested." The project's own deeper investigation of this exact question (`docs/57-data-chosen-carryover-half-life.md`, `docs/series/ch23-data-chosen-half-life.md`) found pooled inflation of 0.067-0.074 across 11 null cells using a richer analysis model (the full recommended `F1` model with expectancy terms) and a wider AIC grid, and explicitly states that even at its own $N = 70$, $t_{1/2} = 1$ cell, the naive-AIC excess was "not borne out cleanly." Paper 02's single-cell, simpler-model check disagrees with the closest comparable result in the project's own prior work, and the disagreement is not acknowledged. Either `G4` and `G9` need a broader Type I error check matching docs/57's pooled design before the power ranking that currently favors them (`G9` is reported as the single best-performing specification) can be trusted, or paper 02 should adopt the bootstrap-calibrated `F1A0B`/`F1AB` selection from `ch23` instead of the naive AIC selection, which was developed specifically to resolve this problem.

5. **Companion-paper citations need the same fabrication check as paper 01.** [unverified] `@pmsimstats_dgp_architectures2026` and `@pmsimstats_calibration2026` are both cited; verify both exist and match what is claimed of them.

6. **Correction (2026-10-21): this item was wrong, same error as paper 01's original item 4.** `report.Rmd` names ADEMP explicitly and repeatedly: "We embed these three in an ADEMP-structured" (line 268), "the ADEMP reporting framework of Morris, White, and Crowther" (line 410), "ADEMP order" (line 417), "ADEMP-compliant cell-level summaries" (line 2163), and "Morris ADEMP protocol" (line 2252). This gap

does not exist; no action is needed. Left here, struck through in substance rather than deleted, so the correction is visible.

7. **Paper 02's "best method" claim is narrower than it reads, and F1A0B's provenance must not be attributed to paper 02.** [verified this session, see conversation record 2026-10-21] Three related problems, surfaced by discussing paper 01's Study 3 and the project's F1 analysis alongside paper 02:

    - Paper 02's `G1-G9` family never includes an expectancy term (`De`/`bmc:De`). Paper 01's own Study 3 (the 14-analysis comparison, Part 1 above) found that term necessary for robustness to a prognostic biomarker: every analysis that shared the biomarker's baseline association with its off-drug association failed at least one prognostic null cell; only the expectancy-term family held nominal size throughout. Paper 02 has never tested a prognostic-biomarker cell at all, so its recommendation of Exposure-weighted (`G3`/`G8`) as the best method is unverified under exactly the condition paper 01 showed matters most.
    - The project's own named "recommended analysis," F1 (`Y ~ bmc + t + Dbc + bmc:Dbc + De + bmc:De`, CR2 cluster-robust SEs, Satterthwaite df; `CLAUDE.md`), is in effect the synthesis of paper 02's winning carryover representation (`Dbc` is identical to `G3`'s Exposure-weighted predictor) with paper 01's necessary expectancy terms. **Neither paper currently states this synthesis or cites the other for it.** A referee reading both companion papers, and then `CLAUDE.md`'s recommended analysis, would have no way to see that F1 is where their two separate conclusions actually meet.
    - `F1A0B` (the bootstrap-calibrated, data-driven half-life selection developed in `docs/57/ch23` specifically to resolve the Type I error risk in gap 4 above) has **never been tested in paper 02 or cited by it**. It must not be described, in either paper or in any summary of this work, as a conclusion paper 02 reached after investigating `G1-G9` — it wasn't, and doing so would repeat the same kind of misattributed-result problem identified for the 15.4%/6.9% figure in Part 1. If F1A0B is eventually recommended, that claim needs its own test on paper 02's (or a successor paper's) own grid before either paper states it.

Plan

Step	Action	Depends on
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1	Fix the cross-factor correlation bug in `simulation-core.R/implementations/tidyverse/R/functions.R` so a single $c_1 = 0.2$ scales by ρ^{lag} at every lag, matching the documented refined-process spec.	none
2	Add randomized path allocation (replace the deterministic `floor`+remainder split with a `sample()`-based assignment) in the same driver.	none
3	Decide, in coordination with paper 01's step 3, whether to fix non-PD handling (stop vs. repair) for this paper's simulations, or carry it forward as a shared, explicitly documented limitation in both papers.	Paper 01 step 3
4	Re-run the affected cells under the corrected settings from steps 1-3. Scope this to the cells that matter for the headline ranking (`G1-G9` at the Hybrid reference grid) rather than the full 216-cell grid if time-boxing is needed; flag any cell not rerun as a known limitation rather than silently leaving stale numbers in place.	1, 2, 3
5	Add a DGP-specification section/table to paper 02 modeled on paper 01's Table `tab:spec`, naming the refined process explicitly and cross-referencing paper 01 as the paper that establishes and justifies it. This also resolves gap 3.	1, 2, 3 (so the table describes what the rerun code actually does)

Step	Action	Depends on
6	Replace "Architecture A/B" with "covariance moderation"/"mean moderation" throughout (table cells, all figure captions, prose).	none, can run in parallel
7	Expand Block S11 to a pooled multi-cell check matching docs/57's design (multiple true half-lives, larger pooled n), or replace `G4`/`G9` with the bootstrap-calibrated `F1A0B`/`F1AB` method from ch23. The second option avoids re-deriving a size guarantee from scratch but requires implementing the bootstrap procedure in `simulation-core.R` and re-running `G4`/`G9`'s cells.	4 (rerun baseline should be settled first)
8	Verify companion citations (shared with paper 01 step 4).	none
9	Name ADEMP conformance explicitly (shared pattern with paper 01 step 5).	none
10	Full read-through against *Trials* Instructions for Authors once confirmed live.	1-9
11	Add an expectancy-term (`De`/`bmc:De`) variant to the `G1-G9` family, and add at least one prognostic-biomarker null cell to the grid (reusing paper 01 Study 3's cell design), before paper 02's "best method" claim is extended to cover a prognostic biomarker.	4
12	Add `F1A0B` (and/or `F1AB`) as a tested candidate alongside `G1-G9`, run on paper 02's own corrected grid (post steps 1-3). Implement it directly from `docs/57-data-chosen-carryover-half-life.md` Section 4.5 (the parametric null-model bootstrap that repeats the selection step in every resample) and `docs/series/ch23-data-chosen-half-life.md`'s construct box and candidate grid ($\mathcal{T}_a = \{0, 0.25, 0.5, 1, 2\}$ weeks); cite both explicitly in paper 02's Methods as the source of the method, not just of the idea. If F1A0B is recommended, this makes paper 02 itself the source of that conclusion, on its own tested grid, rather than an uncited internal document the recommendation is currently borrowed from without attribution.	1, 2, 3, 11

Step	Action	Depends on
13	Once 11-12 land, state the paper-01/paper-02/F1 synthesis explicitly in both papers: paper 02's Discussion should name `Dbc` as the carryover representation its own comparison selected, now combined with paper 01's expectancy terms into F1; paper 01 should re-verify, at least as a confirmatory check, that its headline refined-process power-vs-half-life results are consistent under F1/F1A0B, not only under `E9`/RH's `lmer`. This is a small addition to paper 01 (already otherwise resolved, Part 1) in addition to a paper 02 change.	12

Paper 02's gap is substantially larger than paper 01's. Steps 1, 2, and 6 are mechanical and low-risk. Step 7 is the long pole: it either requires a materially larger S11 rerun or a new bootstrap-calibration implementation, and the choice determines whether `G4`/`G9` can keep their current "best-performing specification" claim at all.

Part 3: Cross-cutting dependencies

- **Paper 02 cannot credibly claim to use "the canonical DGP" until paper 01's own DGP recommendation is stable.** If paper 01's step 1 (restoring or redirecting the matched-N architecture result) changes any parameter defaults or the stated ceiling values, paper 02's rerun (its step 4) should happen after, not before, that settles, to avoid rerunning twice.
- **The non-PD stop-vs-repair decision (paper 01 step 3, paper 02 step 3) is one decision, not two.** Whatever is decided should apply identically to both papers' code paths, since they currently share the same underlying `functions.R`/`generateData.R` implementations.
- **Terminology consistency.** Once paper 02's "Architecture A/B" labels are retired (step 6), confirm no other document in `docs/series/` or `NOTATION.md` still points to paper 02 using the old labels.
- **Citation verification (paper 01 step 4, paper 02 step 8)** should be done once, covering both papers' reference lists together, since they likely share several companion citations.
- **New finding, out of scope for this session: a pre-existing cross-implementation parity regression, unrelated to today's changes.** [verified 2026-10-21] Running `analysis/scripts/parity/test-parity-extended-tidyverse.R` (which checks numerical agreement between `original-extended` and `tidyverse` under matching seeds) reports 144 cells with a sigma- means mismatch between the two implementations, all under `mean_moderation` architecture. Confirmed via `git stash` that this regression is present

identically with or without this session's code changes (same 144-cell count, same pattern), so it predates this work and is not something introduced here. It is a genuine open correctness question for the `tidyverse/original-extended` parity guarantee the project otherwise relies on, worth a dedicated investigation, but was not pursued further in this session since it sits outside both papers' plans above. Flagging it here so it is not lost.

Part 4: Trials-journal checklist (unverified against the live page)

The following are general BMC *Trials* expectations from prior experience with the journal, not confirmed against its current Instructions for Authors during this session. Confirm each before submission:

- Structured abstract (BMC journals typically cap abstracts around 350 words; *Trials* may have its own structure requirement for methodology articles).
- A data and code availability statement naming the public repository and, if applicable, a release tag or commit hash corresponding to the submitted results, not a moving `main` branch.
- A declarations section (funding, competing interests, ethics/ consent, as applicable; likely "not applicable" for a simulation study, but the journal will expect that stated explicitly rather than omitted).
- Explicit identification of the reporting guideline followed for simulation studies (ADEMP), if *Trials* lists this as an expected guideline for methodology submissions in its current instructions.
- Trial registration: not applicable for a simulation/methods paper, but confirm the journal's submission system allows marking this "not applicable" rather than requiring a registration number.
- Exploratory-study framing: per this project's own established convention, neither paper should describe its simulation grid as pre-registered or pre-specified; confirm final text keeps this framing (both papers appeared to do so as of this review, but this was not re-checked line by line during this session).

Summary

Paper 01, updated 2026-10-21: all five in-repository action items (steps 1, 3, 4, 5, and the correction to the ADEMP finding) are resolved. The matched-N architecture-loss result was formally scoped out of paper 01 (it belongs to the mean-vs-covariance architecture question, not paper 01's actual goal of specifying the best covariance-moderation DGP), with `CLAUDE.md` and `ch07-architectures.md` corrected to stop citing "paper 01"

for it. The package now implements its own stop-on-non-PD recommendation via an opt-in `stopOnNonPD` argument, tested and passing the full suite. Paper 01's own citations were checked and all resolve to real entries. The ADEMP-conformance gap in the original version of this analysis was itself an error; the paper already names ADEMP explicitly. Only step 6 (the live *Trials* Instructions-for-Authors read-through) remains, blocked on access to that page rather than on further work here.

Paper 02 is further out and unchanged by this update: it has a real, numeric deviation from the DGP its companion paper recommends, an under-powered and apparently-contradicted Type I error claim supporting its currently-best-performing specification, and pervasive stale terminology (see Part 2). Paper 02's own companion citations (`@pmsimstats_dgp_architectures2026`, `@pmsimstats_calibration2026`) were confirmed this session to be placeholder `@unpublished` bib entries for companion manuscripts that do not exist as actual report directories — the fabrication-risk check flagged as needed for both papers has therefore been done for paper 01 (clean) and partly done for paper 02 (two citations found dangling; full code-level fabrication check still pending).